



GOA INSTITUTE OF MANAGEMENT

Sanquelim Campus, Poriem, Sattari, Goa

Program: BDA/IDA	Exam: Quiz 1
Academic year: 2025-26	Weightage: 10%
Term: II	Maximum Marks: 10
Section: A1/B1/C1	Duration: 30 minutes
Course: Machine Learning with Business Applications	Date: 17 th October, 2025
Name of Instructor: Dr. S. Sanyal	

Instructions:

1. The quiz contains 20 Multiple Choice Questions. Each question is worth 0.5 mark.
2. The quiz is closed book/notes/laptop/internet/etc.
3. Use of calculator is allowed; **calculator exchange is not allowed.**
4. The space for rough work is marked; **additional pages for rough work are not allowed.**
5. Box one correct option from (a), (b), (c), (d), or (e). For example, if you think the correct option for a question is (b), you should mark ☐ (b) ☐

Space for Rough Work

1. In a typical split, the test set is used to:
 - (A) Tune hyperparameters repeatedly
 - (B) Provide the final, unbiased performance estimate
 - (C) Increase training set size to reduce variance
 - (D) Decide which features to include
 - (E) None of options (A), (B), (C), or (D) is correct.
2. A retailer wants to discover natural customer groups for targeted campaigns with no labels available. Which core task fits best?
 - (A) Classification
 - (B) Regression
 - (C) Clustering
 - (D) Recommendation
 - (E) None of options (A), (B), (C), or (D) is correct.
3. A large e-commerce marketplace onboards 5,000 new sellers a week. True fraud cases are rare and evolve quickly. You have listing text, image hashes, price deviations vs. peers, account age, return rates, and network links (shared addresses, phones). Ops wants an automated first-pass screen to catch unusual, high-risk sellers/listings early, then hand off to analysts. What's the primary task?
 - (A) Classification, Prediction
 - (B) Anomaly detection, Prediction
 - (C) Clustering (unsupervised segmentation)
 - (D) Ranking (top- K sellers to review), Prediction
 - (E) None of options (A), (B), (C), or (D) is correct.
4. Which practice most clearly causes optimistically biased model selection?
 - (A) Standardizing using stats computed on the training fold only
 - (B) Reporting test metrics once at the end
 - (C) Stratified sampling to balance classes in folds
 - (D) Selecting features using all rows before the train/validation/test split
 - (E) None of options (A), (B), (C), or (D) is correct.
5. Which statement best captures the accuracy–interpretability trade-off for business ML?
 - (A) Simpler models are easier to explain but may sacrifice some accuracy
 - (B) Complex models are always both more accurate and more interpretable
 - (C) Interpretability and accuracy are unrelated
 - (D) Complex models are never allowed in regulated industries
 - (E) None of options (A), (B), (C), or (D) is correct.
6. Adding an unhelpful predictor to your sales model will most likely do what to adjusted R^2 ?
 - (A) Always increase it
 - (B) Leave it unchanged

- (C) Potentially decrease it
 - (D) Force R^2 to zero
 - (E) None of options (A), (B), (C), or (D) is correct.
7. For a fixed media-mix point, the 95% prediction interval for next city's sales vs the 95% confidence interval for mean sales is typically:
- (A) Prediction interval is wider
 - (B) Prediction interval is narrower
 - (C) Identical widths
 - (D) Unrelated to residual variance
 - (E) None of options (A), (B), (C), or (D) is correct.
8. In logistic regression, e^{β_j} is interpreted as the:
- (A) Multiplicative change in odds per unit of X_j (holding others fixed)
 - (B) Additive change in probability per unit of X_j
 - (C) Additive change in log-odds per unit of X_j
 - (D) Multiplicative change in probability per unit of X_j
 - (E) None of options (A), (B), (C), or (D) is correct.
9. On a perfectly separable dataset, MLE fitting of logistic regression can:
- (A) Produce coefficients that diverge to very large magnitudes (no finite MLE)
 - (B) Force probabilities to 0.5 for all points
 - (C) Make the log-likelihood non-convex
 - (D) Guarantee zero test error in practice
 - (E) None of options (A), (B), (C), or (D) is correct.
10. A video app with 50000 titles needs a daily personalized Top-10 rail per user to maximize watch time. You have user-item interactions, sparse ratings, content metadata, and recent session signals. Product cares about who sees what (top- K), in what order, more than perfectly calibrated probabilities. What's the primary task?
- (A) Clustering (unsupervised audience discovery)
 - (B) Recommendation + Ranking, Prediction
 - (C) Regression, Inference
 - (D) Dimensional reduction (feature compression), unsupervised
 - (E) None of options (A), (B), (C), or (D) is correct.
11. When comparing regression models on the same dataset, which metric is most appropriate if the business wants a single-number summary that penalizes large errors more than small errors and is expressed in the same units as the target?
- (A) R^2
 - (B) Mean Absolute Error (MAE)
 - (C) Root Mean Squared Error (RMSE)
 - (D) Mean Absolute Percentage Error (MAPE)
 - (E) None of options (A), (B), (C), or (D) is correct.

12. A hospital network wants to flag patients likely to be readmitted within 30 days (Yes/No) at discharge to allocate care-manager time. Costs of missing a true readmission are far higher than unnecessary outreach. The team will tune thresholds and produce a ranked list for daily rounds. What is the primary task?
- (A) Classification, Prediction
 - (B) Ranking only, Prediction (no underlying classifier needed)
 - (C) Regression, Inference
 - (D) Anomaly detection, Prediction
 - (E) None of options (A), (B), (C), or (D) is correct.
13. Marketing asks two things: (i) forecast next-month leads, (ii) explain which channels drive leads. Which pairing is most appropriate?
- (A) (i) Inference, (ii) Prediction
 - (B) (i) Prediction, (ii) Inference
 - (C) (i) Clustering, (ii) Prediction
 - (D) (i) Inference, (ii) Clustering
 - (E) None of options (A), (B), (C), or (D) is correct.
14. A supermarket chain has two goals for the next quarter. (i) Forecast next-week SKU-store demand for replenishment. (ii) Quantify the average uplift from price promotions by region, controlling for seasonality and store mix, to guide future spend. Which pairing best matches the two distinct objectives?
- (A) (i) Regression, Prediction (ii) Regression, Inference
 - (B) (i) Classification, Prediction (ii) Clustering (unsupervised)
 - (C) (i) Clustering (unsupervised) (ii) Dimensional reduction (unsupervised)
 - (D) (i) Regression, Inference (ii) Classification, Prediction
 - (E) None of options (A), (B), (C), or (D) is correct.
15. With an intercept and a categorical variable having G regions, how many dummies do you include to avoid the dummy-variable trap? (*The dummy variable trap refers to a situation in regression analysis where including dummy variables for all categories of a categorical predictor, along with an intercept term, results in perfect multicollinearity. This happens because the dummy variables for all categories sum to 1 for every observation, mirroring the intercept, making the model matrix singular and preventing unique estimation of coefficients.*)
- (A) G
 - (B) $G - 1$
 - (C) 1
 - (D) $G + 1$
 - (E) None of options (A), (B), (C), or (D) is correct.
16. With two predictors, the logistic regression decision boundary occurs when:
- (A) $p(x) = 0$
 - (B) $p(x) = 1$
 - (C) $\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$ (log-odds = 0)

- (D) $x_1 = x_2$
- (E) None of options (A), (B), (C), or (D) is correct.
17. In a linear model with predictors **Price** (continuous), **Promo** (0/1; baseline = 0), the coefficient of the interaction **Price:Promo** is best read as:
- (A) Change in the intercept when promotion is on
- (B) Average discount during promotion
- (C) Correlation between price and promotion
- (D) Change in the slope of Price when promotion is on
- (E) None of options (A), (B), (C), or (D) is correct.
18. Your ads platform has ~300 engineered features (user, creative, context). Executives want a 10-factor dashboard that captures most variance and faster downstream modeling; you'll later feed these factors to a propensity model. The immediate, primary need is to compress and denoise the feature space while retaining structure. What's the primary task?
- (A) Classification, Prediction
- (B) Regression, Inference
- (C) Recommendation + Ranking, Prediction
- (D) Dimensional reduction (unsupervised)
- (E) None of options (A), (B), (C), or (D) is correct.
19. A national retailer has millions of basket lines and wants to discover shopper segments (mission shoppers, premium health, bargain bulk) and group SKUs that co-occur for modular layouts. High-dimensional sparse transactions make direct visualization hard, so they'll first compress signals before segmenting. What's the best description of the primary task (and helpful support)?
- (A) Regression, Prediction (primary)
- (B) Classification, Inference (primary)
- (C) Clustering (unsupervised segmentation, primary) + Dimensional reduction (supporting)
- (D) Anomaly detection (primary) + Recommendation (supporting)
- (E) None of options (A), (B), (C), or (D) is correct.
20. In a standard train/validation/test process, what is the primary role of the validation set?
- (A) Final unbiased performance reporting
- (B) Extra training data to reduce variance
- (C) Calibrating probabilities after test
- (D) Hyperparameter tuning/model choice before touching the test set
- (E) None of options (A), (B), (C), or (D) is correct.



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Program: BDA/IDA	Exam: Quiz 2
Academic year: 2025-26	Weightage: 10%
Term: II	Maximum Marks: 10
Section: A1/B1/C1	Duration: 30 minutes
Course: Machine Learning with Business Applications	Date: 28 th November, 2025
Name of Instructor: Dr. S. Sanyal	

Instructions:

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5. Box one correct option from (a), (b), (c), (d), or (e). For example, if you think the correct option for a question is (b), you should mark

(b)

NAME _____

ROLL NUMBER _____

SECTION _____

Space for Rough Work

1. A supply-chain manager applies hierarchical clustering with single linkage to group suppliers. The dendrogram suggests “chaining” effects, where distant suppliers get gradually linked via intermediate ones. Which change is most likely to reduce this chaining?
 - (a) Switch to complete linkage.
 - (b) Switch to using Manhattan distance with single linkage.
 - (c) Increase the number of suppliers in the dataset.
 - (d) Normalize all variables to the range $[0, 1]$ but keep single linkage.
 - (e) None of options (a), (b), (c), or (d) is correct.
2. In hierarchical clustering of product sales patterns, management wants clusters where any two products in the same cluster are relatively similar (tight clusters). Which linkage is most aligned with this requirement?
 - (a) Single linkage.
 - (b) Complete linkage.
 - (c) Average linkage.
 - (d) Centroid linkage with no scaling.
 - (e) None of options (a), (b), (c), or (d) is correct.
3. A data scientist performs hierarchical clustering on standardized customer features and obtains a dendrogram. Management wants exactly 5 customer segments. What is the usual way to obtain these?
 - (a) Choose the 5 shortest branches in the dendrogram.
 - (b) Choose the 5 points closest to the root and form clusters around them.
 - (c) Take the 5 highest points in the dendrogram and treat them as centroids.
 - (d) Cut the dendrogram at the height that yields 5 clusters.
 - (e) None of options (a), (b), (c), or (d) is correct.
4. Which of the following cluster evaluation measures is an *external* measure that requires true class labels?
 - (a) Silhouette Score.
 - (b) Dunn Index.
 - (c) Gap Statistic.
 - (d) Within-Cluster Sum of Squares (WSS).
 - (e) None of options (a), (b), (c), or (d) is correct.
5. A data science team uses k-means to cluster customers and wants to assess how well-separated the clusters are without using external labels. Which measure is most appropriate?
 - (a) Accuracy.
 - (b) Precision.
 - (c) Silhouette score.
 - (d) ROC-AUC.
 - (e) None of options (a), (b), (c), or (d) is correct.

6. What is the primary goal of Principal Component Analysis (PCA)?
 - (a) To create decision trees for classification.
 - (b) To assign class labels to new observations.
 - (c) To reduce the dimensionality of data while retaining most of the variance.
 - (d) To detect outliers in time series only.
 - (e) None of options (a), (b), (c), or (d) is correct.
7. An analytics team compresses 100-dimensional transaction features to 5 principal components using PCA, then trains a classification model and evaluates it with 10-fold Cross-Validation. What is the correct way to integrate PCA into the CV process?
 - (a) Compute PCA once on the full dataset and reuse the components in each fold.
 - (b) Compute PCA separately inside each training fold and apply the resulting transform to its test fold.
 - (c) Compute PCA only on the test folds.
 - (d) Skip PCA inside CV and only apply it after the model is finalized.
 - (e) None of options (a), (b), (c), or (d) is correct.
8. In a customer churn prediction problem, you use Cross-Validation to select the best model and then want an unbiased estimate of its performance. Which practice is most appropriate?
 - (a) Use the same cross-validation scores used for model selection as the final performance estimate.
 - (b) Evaluate the chosen model only on the training folds.
 - (c) Report the highest performance score among all tried hyperparameter settings.
 - (d) After model selection, refit the chosen model on the full training data and evaluate on a separate holdout test set untouched during training and selection.
 - (e) None of options (a), (b), (c), or (d) is correct.
9. In k -fold Cross-Validation, what is the main purpose of using multiple folds instead of a single train-test split?
 - (a) To reduce the size of the dataset.
 - (b) To get a more stable and reliable estimate of model performance.
 - (c) To make the model more complex.
 - (d) To guarantee zero prediction error.
 - (e) None of options (a), (b), (c), or (d) is correct.
10. A marketing analyst applies k-means clustering with Euclidean distance on unscaled features: annual income (in lakhs), number of purchases, and satisfaction score (1–5). What is the main concern with this approach?
 - (a) k-means cannot be used with more than two features.
 - (b) k-means always produces perfectly balanced clusters.
 - (c) The feature with the largest scale (income) will dominate distance calculations.
 - (d) Satisfaction score will dominate because it is ordinal.
 - (e) None of options (a), (b), (c), or (d) is correct.

11. A retail company applies k-means clustering to group customers based on their purchase behaviour. What is the main business benefit of such clustering?
- (a) Ensuring that all customers receive the same promotion.
 - (b) Automatically predicting next quarter's revenue.
 - (c) Creating customer segments to design targeted marketing strategies.
 - (d) Replacing financial forecasting models.
 - (e) None of options (a), (b), (c), or (d) is correct.
12. A PCA on a 40-dimensional customer-features dataset shows that the first 5 principal components explain 65% of the variance, and the first 10 explain 90%. The business wants a compact representation while retaining most information. Which choice is most reasonable?
- (a) Use only the first principal component.
 - (b) Use the first 5 principal components.
 - (c) Use the first 10 principal components.
 - (d) Use all 40 original features.
 - (e) None of options (a), (b), (c), or (d) is correct.
13. A bank analyzes customer data with variables measured in very different units (income, age, credit score, number of products). Before applying PCA, what is usually the most appropriate preprocessing step?
- (a) Standardize each variable (e.g., to zero mean and unit variance).
 - (b) No preprocessing is needed; PCA is scale-invariant.
 - (c) Only center variables without scaling.
 - (d) Normalize each observation vector to have unit length but do not center.
 - (e) None of options (a), (b), (c), or (d) is correct.
14. In a PCA of manufacturing quality data, the first principal component has large positive loadings on "defect rate" and "rework time" and a large negative loading on "first-pass yield." What is the most appropriate interpretation?
- (a) A component capturing overall production volume.
 - (b) A component capturing cost of raw materials.
 - (c) A component capturing a "quality risk" dimension.
 - (d) A component capturing seasonal variation in demand.
 - (e) None of options (a), (b), (c), or (d) is correct.
15. A company wants to compress a high-dimensional feature space and then run k-means clustering on the compressed features. Which pipeline is generally the most appropriate?
- (a) k-means first, then PCA on the cluster labels.
 - (b) PCA first, then k-means on the selected principal components.
 - (c) Run k-means and PCA independently and average the cluster assignments.
 - (d) Perform PCA only on a random subset of points and apply k-means on the full original space.
 - (e) None of options (a), (b), (c), or (d) is correct.

16. In customer segmentation using k-means, what does each cluster centroid most directly represent?
- (a) A randomly chosen customer from that cluster.
 - (b) The customer with the highest revenue in that cluster.
 - (c) The customer closest to the global mean of all customers.
 - (d) A hypothetical “average customer” in that cluster’s feature space.
 - (e) None of options (a), (b), (c), or (d) is correct.
17. In an e-commerce segmentation project, different runs of k-means with the same k but different random initializations give noticeably different clusters. What is the most appropriate response?
- (a) Increase k until clusters stabilize.
 - (b) Decrease k until clusters stabilize.
 - (c) Run k-means multiple times with different initializations and choose the solution with the lowest within-cluster sum of squares.
 - (d) Switch to LOOCV instead of k-means.
 - (e) None of options (a), (b), (c), or (d) is correct.
18. In a credit-risk classification problem with 50,000 loan applications and moderately complex models, which cross-validation strategy offers a good balance between bias, variance, and computation time for performance estimation?
- (a) Leave-One-Out Cross-Validation (LOOCV).
 - (b) 2-fold Cross-Validation.
 - (c) 5- or 10-fold Cross-Validation.
 - (d) Repeated random train–test splits with 90%–10% splits only once.
 - (e) None of options (a), (b), (c), or (d) is correct.
19. In hierarchical clustering, what is a dendrogram?
- (a) A table showing confusion matrix values.
 - (b) A plot of residuals versus fitted values.
 - (c) A line chart of accuracy over epochs.
 - (d) A tree-like diagram that shows how clusters are merged or split at different distances.
 - (e) None of options (a), (b), (c), or (d) is correct.
20. In k-means clustering, what does the parameter k represent?
- (a) The number of features in the dataset.
 - (b) The number of iterations the algorithm will run.
 - (c) The number of clusters to be formed.
 - (d) The number of principal components.
 - (e) None of options (a), (b), (c), or (d) is correct.



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Program: BDA/IDA	Exam: Mid-Term
Academic year: 2025-26	Weightage: 20%
Term: II	Maximum Marks: 50
Section: A1/B1/C1	Duration: 90 minutes
Course: Machine Learning with Business Applications	Date: 29 th October, 2025
Name of Instructor: Dr. S. Sanyal	

Instructions:

1. Write your answer in the space provided.
2. The Mid-Term is closed book/notes/laptop/internet/etc.
3. Use of calculator is allowed; **calculator exchange is not allowed.**
4. The space for rough work is marked; **additional pages for rough work are not allowed.**
5. **Do not use pencil to write your answers.**

Space for Rough Work

Case 1: A health insurer wants to modernize its claims triage and pricing workflow. The insurer operates in three regions with slightly different patient profiles (higher average comorbidities in Region C). They have historical, labeled claims with variables such as patient age, prior conditions, treatment type, hospital tier, billed amount, length of stay, and an audited label **Fraud** (Yes/No). The insurer must keep the model explainable for auditors and hospital partners.

- **Detecting Fraudulent Claims:** Estimate the probability that an incoming claim is fraudulent using features like treatment mix, billing patterns, and length of stay. Use a threshold to classify claims as “*Fraud*” or “*Genuine*” for fast-track/manual review.
- **Estimating Claim Amount Drivers:** Understand how features such as hospital tier, treatment type, and length of stay affect the *expected claim amount* for actuarial pricing and negotiation guidance with providers.

- (a) (2 marks) Referring to Case 1, what is the primary task when deciding whether a claim is fraudulent?

- (b) (2 marks) Referring to Case 1, training on historical labeled claims (Fraud Yes/No) is what type of learning?

- (c) (2 marks) Referring to Case 1, what is the primary task when quantifying how features affect expected claim amount?

- (d) (2 marks) Referring to Case 1, the insurer has 2,400 historical claims and uses 12 features. Report n and p .

- (e) (2 marks) Referring to Case 1, for deployment that screens new claims in real time, is the focus mainly *inference* or *prediction*?

Case 2: A telecom operator wants to reduce monthly churn and improve cross-sell. Behavior drifts across quarters (promotions, tariff changes). Data include recent call usage, data pack purchases, complaints, tenure, region, average revenue per user (ARPU), and a label **Churn** (Yes/No). Management wants a first deployment that is easy to explain, with plans to iterate if accuracy gains are material.

- **Predicting Churn Risk:** Estimate the probability that a subscriber will churn next month, then classify as “*At Risk*” or “*Safe*” to drive retention calls or offers.
- **Understanding Revenue Drivers:** Identify which factors most influence *ARPU* to target cross-sell (e.g., data-add-on packs) and forecast revenue impact by segment.

- (a) (2 marks) Referring to Case 2, what is the primary task when deciding whether a subscriber will churn next month?

- (b) (2 marks) Referring to Case 2, if a complex neural model is chosen to maximize accuracy, what is most likely sacrificed according to the accuracy–interpretability trade-off?

- (c) (2 marks) Referring to Case 2, two candidate churn models have the following out-of-sample properties:

- Model A: $\text{Bias}^2 = 0.0225$, $\text{Variance} = 0.0100$;
- Model B: $\text{Bias}^2 = 0.0025$, $\text{Variance} = 0.0400$.

Assume irreducible noise $\sigma^2 = 0.0200$. Which model has the lower expected test MSE?

- (d) (2 marks) Referring to Case 2, on a 200-subscriber holdout, the model predicted 50 as “At Risk.” Of these 50, 35 actually churned. Among the remaining 150 predicted “Safe,” 20 churned. Report the *accuracy*.

- (e) (2 marks) Referring to Case 2, when modeling ARPU as a function of customer features, which task type is appropriate?

Case 3: A retail platform wants a simple pricing/uplift model for a seasonal promotion. The team collected a small dataset with:

Spend_k (continuous): average weekly promotional spend per customer segment (in \$thousands).

Segment (categorical): Retail or Online.

Revenue_k (response): average weekly revenue per segment (in \$thousands).

The data head and linear model fit are given below.

```
case3 <- read.csv("case3_twofactor_dataset.csv")
head(case3)
```

```
##   id Segment Spend_k Revenue_k
## 1  1  Retail    2.39    31.79
## 2  2  Online    4.78    53.51
## 3  3  Online    7.14    66.31
## 4  4  Retail    1.93    31.30
## 5  5  Online    4.91    54.87
## 6  6  Retail    5.58    40.54
```

```
model_case3 <- lm(Revenue_k ~ Spend_k + Segment, data = case3)
summary(model_case3)
```

```
##
## Call:
## lm(formula = Revenue_k ~ Spend_k + Segment, data = case3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.6331 -1.8449 -0.4271  1.9174  5.0756
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   35.1556    1.0677   32.93  <2e-16 ***
## Spend_k        3.6525    0.1765   20.69  <2e-16 ***
## SegmentRetail -13.9608    0.8870  -15.74  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.771 on 37 degrees of freedom
## Multiple R-squared:  0.9464, Adjusted R-squared:  0.9435
## F-statistic: 326.7 on 2 and 37 DF,  p-value: < 2.2e-16
```

- (a) (2 marks) Referring to Case 3, identify the task type for predicting **Revenue_k** from **Spend_k** and **Segment**.

- (b) (2 marks) Referring to Case 3, using the fitted model, what is the predicted **Revenue_k** for **Retail** when **Spend_k** = 5? (*Round to two decimals.*)

- (c) (2 marks) Referring to Case 3, using the fitted model, find the value of **Spend_k** required (for **Online**) to achieve a predicted **Revenue_k** of 60. (*Round to two decimals.*)

- (d) (2 marks) Referring to Case 3, at **Spend_k** = 5, what is the incremental lift (**Online** minus **Retail**) in **Revenue_k**? (*Round to two decimals.*)

- (e) (2 marks) Referring to Case 3, which segment has the higher baseline revenue at **Spend_k** = 0?

Case 4: A telecom operator wants to proactively manage monthly churn during a tariff reshuffle. Historical analysis suggests two practical churn drivers:

Tickets_30d (continuous): number of customer-support tickets raised in the last 30 days (scaled to a 0–10 range with decimals).

Plan (categorical): **Standard** or **Premium**. A Premium plan includes concierge support and priority issue resolution.

To balance speed and transparency for the retention team, the analytics group fits a logistic regression model.

```
# Logistic regression (Churn ~ Tickets_30d + Plan)
# Read and display head of the data set
case4 <- read.csv("case4_churn_dataset.csv", stringsAsFactors = TRUE)
head(case4)
```

```
##   id Tickets_30d   Plan Churn
## 1  1         4.98 Standard   No
## 2  2         9.85 Standard   Yes
## 3  3         4.48 Standard   No
## 4  4         8.42 Standard   Yes
## 5  5         9.29 Standard   Yes
## 6  6         5.53 Premium   Yes
```

```
# Ensure factor coding:
# - Model the probability of "Yes" (success) explicitly
# - Use "Standard" as the baseline for Plan
case4$Churn <- factor(case4$Churn, levels = c("Yes", "No"))
case4$Plan <- relevel(factor(case4$Plan), ref = "Standard")
# Fit logistic regression (logit link is default for binomial)
model_case4 <- glm(Churn ~ Tickets_30d + Plan, data = case4, family = binomial)
# Parameter estimates and inference
summary(model_case4)
```

```
##
## Call:
## glm(formula = Churn ~ Tickets_30d + Plan, family = binomial,
##      data = case4)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   3.0674     0.8022   3.824 0.000131 ***
## Tickets_30d  -0.5399     0.1322  -4.083 4.45e-05 ***
## PlanPremium   1.8949     0.6785   2.793 0.005224 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 103.591  on 79  degrees of freedom
## Residual deviance:  71.622  on 77  degrees of freedom
## AIC: 77.622
##
## Number of Fisher Scoring iterations: 5
```

- (a) (2 marks) Referring to Case 4, compute the log-odds of churn for a Premium subscriber with Tickets_30d=3.5. (Round to two decimals.)

- (b) (2 marks) Referring to Case 4, compute the predicted probability of churn for a **Standard** subscriber with **Tickets_30d**=7.0. (*Round to three decimals.*)

- (c) (2 marks) Referring to Case 4, find the decision boundary (the **Tickets_30d** value where $p=0.5$) for a **Standard** subscriber. (*Round to two decimals.*)

- (d) (2 marks) Referring to Case 4, holding **Plan** fixed, what is the change in the log-odds of churn when **Tickets_30d** increases by +2.0? (*Round to two decimals.*)

- (e) (2 marks) Referring to Case 4, what is the total number of customers in the dataset `case4_churn_dataset.csv`?

Case 5: A bank is piloting an early-warning score for credit card default using Linear Discriminant Analysis (LDA). Analysts assembled a concise dataset with:

Balance_k (continuous): current credit card balance in \$thousands

LatePays (continuous/int): number of late payments in the last 12 months

Default (factor): **Yes** or **No**

The team fits an LDA to classify new customers as likely to default (**Yes**) or not (**No**) based on **Balance_k** and **LatePays**.

```
library(MASS)
credit_data <- read.csv("case5_lda_dataset.csv")
head(credit_data)
```

```
##      id Balance_k LatePays Default
## 1  91   1880.11         4      Yes
## 2 108   2099.34         5      Yes
## 3  37    530.38         2       No
## 4  67    537.19         0       No
## 5  92   1479.64         2      Yes
## 6  24    509.28         0       No
```

```
credit_data$Default <- factor(credit_data$Default, levels = c("No", "Yes"))
lda_model <- lda(Default ~ Balance_k + LatePays, data = credit_data)
print(lda_model)
```

```
## Call:
## lda(Default ~ Balance_k + LatePays, data = credit_data)
##
## Prior probabilities of groups:
##      No  Yes
## 0.75 0.25
##
## Group means:
##      Balance_k LatePays
## No      810.2089 2.766667
## Yes    1695.7347 5.433333
##
## Coefficients of linear discriminants:
##                      LD1
## Balance_k  0.00350708
## LatePays  -0.00679376
```

```
lda_predictions <- predict(lda_model, credit_data)
table(Actual = credit_data$Default, Predicted = lda_predictions$class)
```

```
##      Predicted
## Actual No  Yes
##      No  87   3
##      Yes   2  28
```

- (a) (2 marks) Referring to Case 5, what is the prior probability of No (non-default)?
(Round to two decimals.)

- (b) (2 marks) Referring to Case 5, what is the average Balance_k for Yes (defaulters)?
(Round to two decimals.)

- (c) (2 marks) Referring to Case 5, what is the total number of customers in the dataset `case5_lda_dataset.csv`?

- (d) (2 marks) Referring to Case 5, what is the accuracy of the LDA classifier? (*Round to two decimals.*)

- (e) (2 marks) Referring to Case 5, if the LDA model misclassifies a **Yes** (defaulter) as **No**, what type of error is this? (*Type I/Type II*)

Space for Rough Work

Space for Rough Work



GOA INSTITUTE OF MANAGEMENT

Sanquelim Campus, Poriem, Sattari, Goa

Program: BDA/IDA	Exam: End-Term
Academic year: 2025-26	Weightage: 30%
Term: II	Maximum Marks: 80
Section: A1/B1/C1	Duration: 150 minutes
Course: Machine Learning with Business Applications	Date: 8 th December, 2024
Name of Instructor: Dr. S. Sanyal	

Instructions:

1. Write your answer in the space provided.
2. The End-Term is closed book/notes/laptop/internet/etc.
3. Use of calculator is allowed; **calculator exchange is not allowed**.
4. The space for rough work is marked; **additional pages for rough work are not allowed**.
5. Unless stated otherwise, round all numerical answers to **two decimal places**.
6. **Rounding Rule (to 2 decimal places):** Look at the **third digit after the decimal**.
 - If it is **0–4**, keep the first two decimal digits unchanged.
 - If it is **5–9**, increase the second decimal digit by 1 (carry to the left if needed).
 - Always write **exactly two digits** after the decimal (use trailing zeros if needed).

Examples:

- **Already exactly 2 decimals (no rounding needed):** 12.30 $\rightarrow$ 12.30, 0.05 $\rightarrow$ 0.05
- **Fewer than 2 decimals (pad zeros):** 7 $\rightarrow$ 7.00, 4.5 $\rightarrow$ 4.50, -3.1 $\rightarrow$ -3.10
- **More than 2 decimals, round down (third digit 0–4):** 3.141 $\rightarrow$ 3.14, 2.004 $\rightarrow$ 2.00, -8.234 $\rightarrow$ -8.23
- **More than 2 decimals, round up (third digit 5–9) without carry:** 3.145 $\rightarrow$ 3.15, 6.629 $\rightarrow$ 6.63, -1.235 $\rightarrow$ -1.24
- **Round up with carry in the hundredths place (e.g., 0.099 $\rightarrow$ 0.10):** 1.095 $\rightarrow$ 1.10, -2.995 $\rightarrow$ -3.00
- **Round up with carry across the integer boundary (e.g., 9.999 $\rightarrow$ 10.00):** 9.999 $\rightarrow$ 10.00, 99.995 $\rightarrow$ 100.00
- **Leading zeros / very small magnitudes:** 0.0004 $\rightarrow$ 0.00, 0.005 $\rightarrow$ 0.01, -0.0049 $\rightarrow$ -0.00 (write as 0.00 if your convention avoids -0.00)

Space for Rough Work

Case 1: A food-delivery platform wants to design “Frequently Bought Together” bundles and optimize add-on recommendations at checkout. The platform collected transactional data from 10 orders. Using the Apriori algorithm, association rules were derived to uncover relationships between frequently purchased items such as Burger, Fries, Coke, IceCream, and Salad.

```
library(arules)
items <- c(
  "Burger", "Fries", "Coke",           # T1 (3)
  "Burger", "Fries", "Coke", "IceCream", # T2 (4)
  "Burger", "Fries",                   # T3 (2)
  "Fries",                             # T4 (1)
  "Burger", "Salad",                   # T5 (2)
  "Burger", "Fries", "IceCream",       # T6 (3)
  "Salad", "Coke",                     # T7 (2)
  "Burger",                           # T8 (1)
  "Salad", "IceCream",                 # T9 (2)
  "Salad", "Fries", "Coke"             # T10 (3)
)
tid <- rep(1:10, times = c(3,4,2,1,2,3,2,1,2,3))
transactions <- as(split(items, tid), "transactions")
inspect(transactions)
```

```
##      items      transactionID
## [1] {Burger, Coke, Fries}      1
## [2] {Burger, Coke, Fries, IceCream} 2
## [3] {Burger, Fries}           3
## [4] {Fries}                   4
## [5] {Burger, Salad}           5
## [6] {Burger, Fries, IceCream}  6
## [7] {Coke, Salad}             7
## [8] {Burger}                   8
## [9] {IceCream, Salad}          9
## [10] {Coke, Fries, Salad}     10
```

```
# Perform Apriori analysis
invisible(capture.output({
rules <- apriori(transactions,
parameter = list(supp = 0.25, conf = 0.4, target = "rules"))}))
# View the rules
inspect(rules)
```

```
##      lhs      rhs      support confidence coverage lift      count
## [1] {}      => {Salad} 0.4      0.4000000    1.0      1.000000 4
## [2] {}      => {Coke}  0.4      0.4000000    1.0      1.000000 4
## [3] {}      => {Burger} 0.6      0.6000000    1.0      1.000000 6
## [4] {}      => {Fries} 0.6      0.6000000    1.0      1.000000 6
## [5] {Coke}   => {Fries} 0.3      0.7500000    0.4      1.250000 3
## [6] {Fries}  => {Coke}  0.3      0.5000000    0.6      1.250000 3
## [7] {Burger} => {Fries} 0.4      0.6666667    0.6      1.111111 4
## [8] {Fries}  => {Burger} 0.4      0.6666667    0.6      1.111111 4
```

1. (10 marks) Answer the questions below by referring to Case 1.

The product team wants the single most reliable "attach" rule to trigger a *combo suggestion* at checkout. Among the non-empty antecedent rules, which rule has the **highest confidence**?

The finance team only approves recommendations that increase purchase likelihood by at least 20% above baseline. Which rule meets this policy?

The growth team is deciding whether to recommend **Fries** to a customer who already has **Coke**. What is the model's estimated **probability** of Fries given Coke? (Round to **two decimals**.)

A business analyst claims: "*Coke and Fries are mutually reinforcing add-ons.*" Which single metric directly supports this claim by showing both directions improve over baseline?

The CFO will approve only one cross-sell nudge, and wants the one that maximizes *incremental conversion uplift* over the baseline popularity of the recommended item. Using the rules table, choose the rule that maximizes:

$$\text{Uplift} = \text{confidence} - \text{support}(\text{rhs}).$$

Which **rule number** should be selected?

Case 2: A BPO wants to cluster agents based on Average Handling Time (AHT) in minutes, for 6 agents [120, 128, 140, 205, 215, 405]. Euclidean distance is used (so distance is absolute difference here, since it's 1D). The firm compares

- Single Linkage: To form elongated clusters and study customer spending chains.
- Complete Linkage: To form compact and tightly grouped clusters.

A dendrogram is generated to visualize the hierarchical structure, and the number of clusters is determined by cutting the dendrogram at a specific height.

2. (10 marks) Answer the questions below by referring to Case 2.

Under complete linkage, if you cut the dendrogram at height 15, how many clusters form?

In the final step of complete linkage, which two clusters merge (describe them by their member values)?

Under single linkage, which two agents (values) merge first?

If the goal is high within-cluster similarity, which linkage should be preferred?

At what height does this first merge occur in single linkage?

Case 3: A **telecom company** wants to reduce network congestion by classifying cell towers into two operational segments using **k-means clustering** with $k = 2$. The segmentation will be used to plan capacity upgrades and preventive maintenance:

- **Hotspot Towers:** towers with unusually high peak-hour traffic (likely to face congestion).
- **Normal-load Towers:** towers with moderate peak-hour traffic.

The telecom team measures **peak-hour average data traffic (in TB/day)** for six towers:

Tower	T1	T2	T3	T4	T5	T6
Peak Traffic (TB/day)	8	10	12	28	30	33

The initial centroids for the two clusters are set to **8 (Normal-load)** and **28 (Hotspot)**. The standard k-means steps are followed:

Step 1: Assign each tower to the nearest centroid using absolute distance (1D Euclidean).

Step 2: Update each centroid as the mean of points in its cluster.

Step 3: Repeat assignment and centroid update until centroids stabilize.

3. (10 marks) Answer the questions below by referring to Case 3.

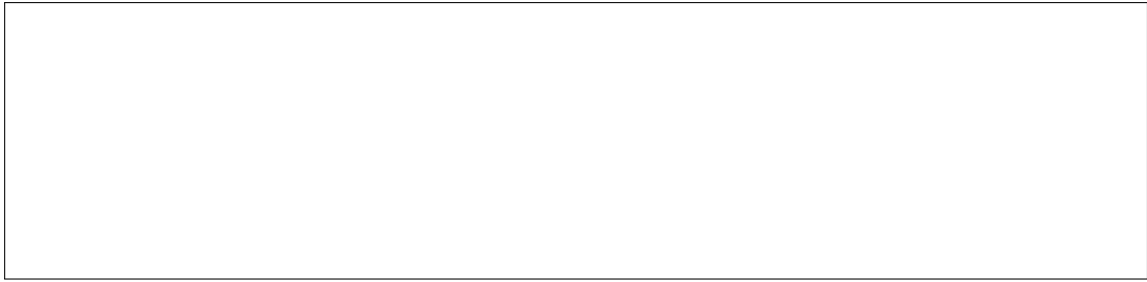
After the **first assignment step** (using centroids 8 and 28), list the towers assigned to the **Hotspot** cluster.

Compute the **updated centroids** after the first iteration (report both centroids; round to **two decimals**.)

Report the **final centroids** after convergence. (Round to **two decimals**.)

The operations head wants a quick congestion rule: classify a newly observed tower as **Hotspot** if its traffic is closer to the Hotspot centroid than the Normal centroid. For a new tower with traffic = **20 TB/day**, which cluster will it be assigned to (based on the final centroids)?

From a business decision perspective, the CFO asks: “Which segmentation outcome is more actionable?” Choose one: **(i)** clusters that are highly balanced in size, or **(ii)** clusters that isolate a small set of extreme towers.



Case 4: A hospital network wants to profile its outpatient clinics to improve **patient flow, staffing, and service quality**. Each clinic is described by four operational attributes:

- **Avg__Wait__Time** (minutes)
- **Patients__Per__Day** (count)
- **Staff__Per__Shift** (count)
- **30Day__Readmission__Rate** (%)

Because the four variables are on different scales and are jointly informative, the network uses **Principal Component Analysis (PCA)** to:

- reduce the 4D operational space to **2 principal components** while retaining most information,
- visualize clinics in a **2D map** to spot operational clusters/outliers,
- interpret which operational drivers dominate each principal component.

The data for five clinics is shown below:

Clinic	Avg__Wait__Time	Patients__Per__Day	Staff__Per__Shift	30Day__Readmission__Rate
A	18	120	10	9.0
B	25	95	9	12.0
C	35	140	11	15.0
D	15	80	8	7.5
E	40	160	12	16.5

```
# Load necessary libraries
library(ggplot2)
# Data: outpatient clinic operations
data <- data.frame(
  Avg_Wait_Time = c(18, 25, 35, 15, 40),           # minutes
  Patients_Per_Day = c(120, 95, 140, 80, 160),      # count
  Staff_Per_Shift = c(10, 9, 11, 8, 12),            # count
  Readmission_30D = c(9.0, 12.0, 15.0, 7.5, 16.5)   # percent
)
# Clinic names
clinic_names <- c("A", "B", "C", "D", "E")
# Standardize the data
```

```
data_scaled <- scale(data)
# Perform PCA (prcomp centers by default; scale.=TRUE ensures standardization)
pca_result <- prcomp(data_scaled, scale. = TRUE)
# Summary of PCA
summary(pca_result)
```

```
## Importance of components:
##              PC1      PC2      PC3      PC4
## Standard deviation    1.9309 0.51632 0.07084 0.01267
## Proportion of Variance 0.9321 0.06665 0.00125 0.00004
## Cumulative Proportion 0.9321 0.99871 0.99996 1.00000
```

```
# Loadings (eigenvectors)
loadings <- pca_result$rotation
print(loadings)
```

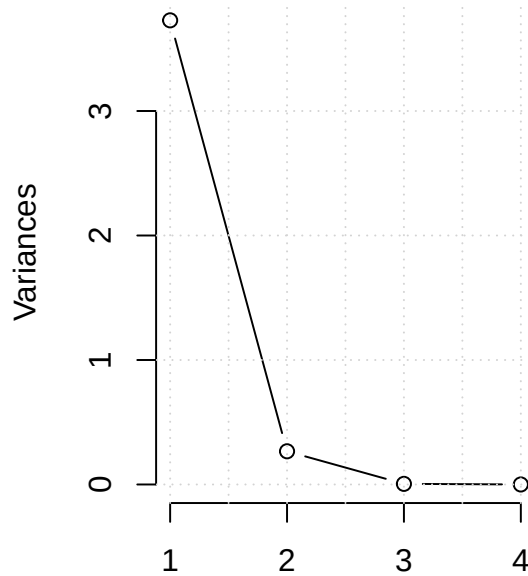
```
##              PC1      PC2      PC3      PC4
## Avg_Wait_Time    0.5022180 -0.4643223 0.6564989 0.3181055
## Patients_Per_Day 0.4974572 0.5370719 0.2964942 -0.6133362
## Staff_Per_Shift  0.5028914 0.4599296 -0.3663793 0.6335071
## Readmission_30D  0.4974070 -0.5333136 -0.5889538 -0.3482762
```

```
# Scores (clinic coordinates in PC space)
pca_scores <- pca_result$x
rownames(pca_scores) <- clinic_names
print(pca_scores)
```

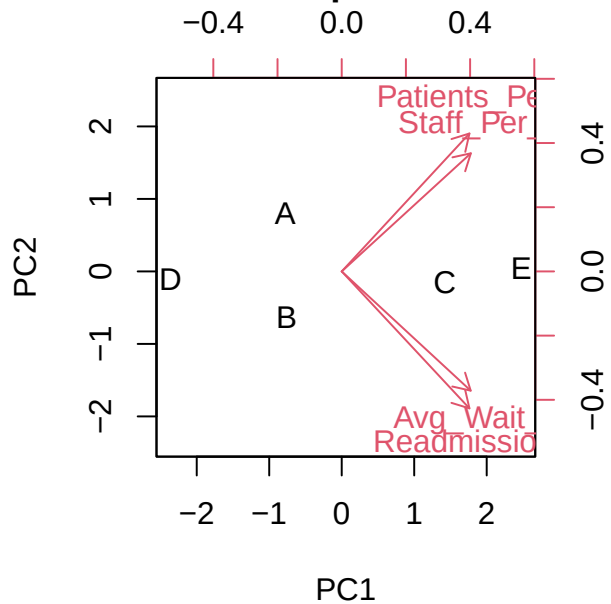
```
##              PC1      PC2      PC3      PC4
## A -0.7771142 0.80678112 -0.05465495 -0.0004466116
## B -0.7604608 -0.61853975 -0.08518325 0.0051286171
## C 1.4227561 -0.14347607 0.01153115 -0.0202406559
## D -2.3612570 -0.09748337 0.09123947 0.0012762899
## E 2.4760759 0.05271808 0.03706759 0.0142823605
```

```
# Plot: Scree plot and biplot side by side
par(mfrow = c(1, 2))
# Scree Plot
plot(pca_result, type = "l", main = "Scree Plot")
grid()
# Biplot
biplot(pca_result, scale = 0, main = "Biplot",
xlabs = clinic_names)
```


Scree Plot



Biplot



4. (10 marks) Answer the questions below by referring to Case 4.

How many principal components are needed to retain at least **95%** of the total variance?

What percentage of total variance is **not** explained by the first two components (PC1 and PC2)? Express your answer as a percentage (round to **two decimals**).

Using the clinic scores, which clinic is the **most extreme high-intensity clinic** and which is the **most extreme low-intensity clinic** along PC1? (Name both clinics.)

Consider the 2D map using only PC1 and PC2. Compute the Euclidean distance between clinics **C** and **E** using their (PC1, PC2) scores, and comment briefly on whether they seem **similar or dissimilar** in the PCA space.

Which principal component represents a "high-throughput, well-staffed, efficient / better-outcomes" operational profile?



Case 5: An airport ground-handling company wants to reduce operational penalties caused by baggage delays. Management believes the penalty cost is driven primarily by:

- **Flights Handled** (count per day)
- **Weather Disruption Index** (0 = clear, 5 = severe disruption)
- **Baggage Delay Penalty Cost** (in lakhs per day)

The company builds a **linear regression** model to:

- quantify how Flights and Weather jointly affect penalty cost,
- predict penalty cost for future operating conditions,
- evaluate model performance using **Leave-One-Out Cross-Validation (LOOCV)**.

Day	Flights	Weather Index	Penalty Cost (lakhs)
1	120	2.0	6.8
2	95	3.5	6.5
3	140	2.5	7.4
4	80	4.0	6.0
5	160	1.8	8.3
6	110	3.0	6.2

```
# Load the required library for cross-validation and model training
library(caret)
# Define the dataset
data <- data.frame(
  Flights = c(120, 95, 140, 80, 160, 110),      # flights handled per day
  Weather = c(2.0, 3.5, 2.5, 4.0, 1.8, 3.0),    # disruption index (0 to 5)
  PenaltyCost = c(6.8, 6.5, 7.4, 6.0, 8.3, 6.2) # lakhs per day
)
# LOOCV setup: each day acts as test once
train_control <- trainControl(method = "LOOCV", savePredictions = "final")
# Train linear regression with LOOCV
cv_model <- train(PenaltyCost ~ Flights + Weather,
  data = data,
  method = "lm",
  trControl = train_control)
# Cross-validated metrics (RMSE, R-squared, MAE)
print("Cross-validated results:")
```

```
## [1] "Cross-validated results:"
```

```
print(cv_model$results)
```

```
##   intercept      RMSE Rsquared      MAE
## 1      TRUE 0.4843613 0.6909671 0.453438
```

```
# Final model coefficients (fit on full data)
print("Final model coefficients:")
```

```
## [1] "Final model coefficients:"
```

```
print(coef(cv_model$finalModel))
```

```
## (Intercept)      Flights      Weather
##  2.20000140  0.03399463  0.24010565
```

```
# LOOCV predictions + errors
pred_df <- cv_model$pred
pred_df$AbsError <- abs(pred_df$pred - pred_df$obs)
pred_df <- pred_df[order(-pred_df$AbsError), ]
print("LOOCV predictions (sorted by absolute error):")
```

```
## [1] "LOOCV predictions (sorted by absolute error):"
```

```
print(pred_df)
```

```
##   intercept      pred obs rowIndex  AbsError
## 1      TRUE 6.096938 6.8         1 0.7030620
## 5      TRUE 7.713004 8.3         5 0.5869955
## 6      TRUE 6.760500 6.2         6 0.5604999
## 2      TRUE 6.171039 6.5         2 0.3289612
## 3      TRUE 7.670556 7.4         3 0.2705556
## 4      TRUE 5.729446 6.0         4 0.2705536
```

5. (10 marks) Answer the questions below by referring to Case 5.

(a) Which predictor has a **larger marginal impact** on daily penalty cost: **Flights** or **Weather**?

Predict the penalty cost (lakhs) for a day with **Flights = 130** and **Weather = 2.8**. (Round to **two decimals**.)

From the LOOCV table (sorted by absolute error), which day has the **largest absolute prediction error**? Report the day number and the error value.

The CFO asks: "On average, how far off (in lakhs) are our LOOCV predictions of daily baggage-delay penalty cost?" What would you answer?

What **percentage of variation** in penalty cost is explained by the model?

Case 6: A ride-hailing platform wants to predict whether a driver will **churn in the next 30 days** (Yes/No) so that the operations team can proactively offer retention incentives. The dataset is imbalanced because most drivers do **not** churn. The platform uses:

- **Cancellations** (number of rider cancellations attributed to the driver in the last 30 days)
- **Weekly_Trips** (average trips per week)
- **Avg_Rating** (average driver rating out of 5)
- **Verified** (whether the driver has completed full KYC verification: Yes/No)

To handle class imbalance, **SMOTE** is applied to balance the classes. A **logistic regression** model is then fit to estimate churn probability. Performance is evaluated using:

- Confusion matrix (at threshold 0.5),
- ROC curve and AUC.

```
# Libraries
library(ROSE) # SMOTE / ovun.sample
library(pROC) # ROC + AUC
set.seed(123)
# -----
# Simulated imbalanced churn dataset
# -----
n <- 2500
```

```

data <- data.frame(
  Cancellations = rpois(n, lambda = 1.3),
  Weekly_Trips = pmax(0, round(rnorm(n, mean = 28, sd = 10))),
  Avg_Rating = pmin(5, pmax(3, rnorm(n, mean = 4.55, sd = 0.25))),
  Verified = factor(ifelse(runif(n) < 0.75, "Yes", "No"))
)
# Create churn probability (logistic) so churn is relatively rare (~8-12%)
linpred <- -4.2 +
0.55 * data$Cancellations +
(-0.04) * data$Weekly_Trips +
(-1.10) * (data$Avg_Rating - 4.5) +
(-0.60) * (data$Verified == "Yes")
prob_churn <- 1 / (1 + exp(-linpred))
data$Churn <- factor(ifelse(runif(n) < prob_churn, "Yes", "No"))
# Check class imbalance
print("Class distribution (original):")

```

```
## [1] "Class distribution (original):"
```

```
print(table(data$Churn))
```

```
##
##   No   Yes
## 2476   24
```

```

# -----
# SMOTE to balance the dataset
# -----
# Make N roughly equal to 2 * majority-class count
N_target <- 2 * max(table(data$Churn))
smote_data <- ovun.sample(
  Churn ~ Cancellations + Weekly_Trips + Avg_Rating + Verified,
  data = data,
  method = "both",
  N = N_target,
  seed = 123
)$data
print("Class distribution (after SMOTE):")

```

```
## [1] "Class distribution (after SMOTE):"
```

```
print(table(smote_data$Churn))
```

```
##
##   No   Yes
## 2496 2456
```

```

# -----
# Logistic regression
# -----
model <- glm(Churn ~ Cancellations + Weekly_Trips + Avg_Rating + Verified,
  data = smote_data, family = binomial)
print("Summary of Logistic Regression Model:")

```

```
## [1] "Summary of Logistic Regression Model:"
```

```
print(summary(model))
```

```
##
## Call:
## glm(formula = Churn ~ Cancellations + Weekly_Trips + Avg_Rating +
##       Verified, family = binomial, data = smote_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   3.909728   0.586856   6.662 2.70e-11 ***
## Cancellations  0.678051   0.027276  24.859 < 2e-16 ***
## Weekly_Trips  -0.096242   0.003997 -24.078 < 2e-16 ***
## Avg_Rating     -0.501443   0.127053  -3.947 7.92e-05 ***
## VerifiedYes    -0.722871   0.072701  -9.943 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6864.6  on 4951  degrees of freedom
## Residual deviance: 5235.2  on 4947  degrees of freedom
## AIC: 5245.2
##
## Number of Fisher Scoring iterations: 4
```

```
# Predicted probabilities on the (SMOTE-balanced) data
predicted_probs <- predict(model, type = "response")
# Class prediction with threshold 0.5
predicted_classes <- ifelse(predicted_probs > 0.5, "Yes", "No")
# Confusion matrix
confusion_matrix <- table(Predicted = predicted_classes, Actual = smote_data$Churn)
print("Confusion Matrix (threshold = 0.5):")
```

```
## [1] "Confusion Matrix (threshold = 0.5):"
```

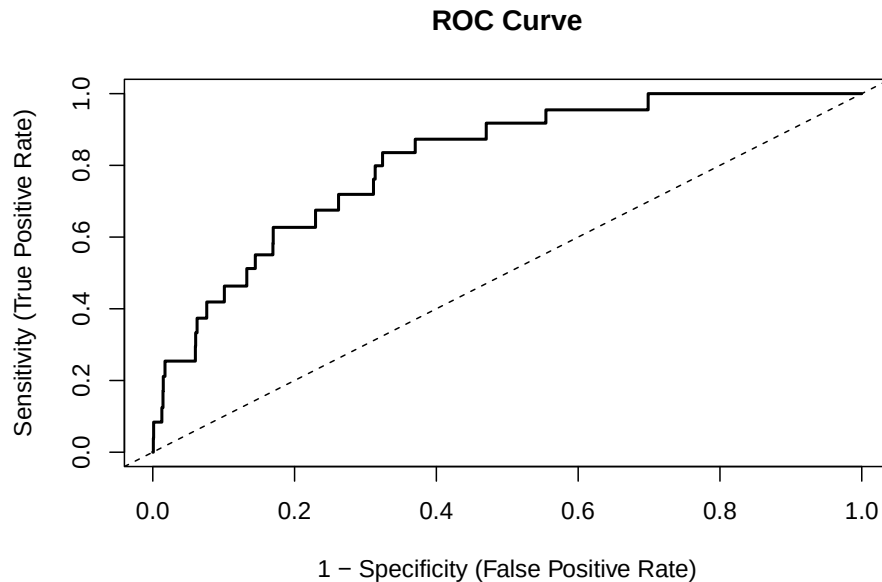
```
print(confusion_matrix)
```

```
##           Actual
## Predicted  No  Yes
##      No  1840  690
##      Yes   656 1766
```

```
# ROC + AUC (set levels so 'Yes' is treated as positive class)
roc_obj <- roc(smote_data$Churn, predicted_probs, levels = c("No", "Yes"), direction = "<")
auc_value <- auc(roc_obj)
print(paste("AUC:", round(auc_value, 3)))
```

```
## [1] "AUC: 0.811"
```

```
# Plot ROC
plot(1 - roc_obj$specificities, roc_obj$sensitivities, type = "l",
     xlab = "1 - Specificity (False Positive Rate)",
     ylab = "Sensitivity (True Positive Rate)",
     main = "ROC Curve", lwd = 2)
abline(a = 0, b = 1, lty = 2)
```



6. (10 marks) Answer the questions below by referring to Case 6.

After SMOTE, the analytics lead asks how balanced the training set became. What is the **post-SMOTE churn proportion** (Yes as a %), round to **two decimals**?

For targeting interventions, which predictor is the strongest **risk-increasing driver** of churn?

In operations terms, how many cases are **false alarms** at threshold 0.5?

For judging ranking quality of the churn model, what is the **Area Under the Receiver Operating Characteristic Curve** (round to **two decimals**)?

After fitting the churn model using Cancellations, Weekly_Trips, Avg_Rating, and Verified, the risk team asks: “What is the model’s **remaining lack-of-fit** after accounting for all predictors?” What would be your answer?

Case 7: Refer to the output below.

```
# Load data
credit_data <- read.csv("Credit.csv")
# Fit the multiple linear regression model
credit_model_reduced <- lm(Balance ~ Income + Limit + Rating + Cards + Age + Student,
                           data = credit_data)
# Model summary
summary(credit_model_reduced)
```

```
##
## Call:
## lm(formula = Balance ~ Income + Limit + Rating + Cards + Age +
##      Student, data = credit_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -170.00  -77.85  -11.84   56.87  313.52
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -493.73419    24.82476  -19.889  < 2e-16 ***
## Income       -7.79508     0.23342  -33.395  < 2e-16 ***
## Limit         0.19369     0.03238    5.981 4.98e-09 ***
## Rating        1.09119     0.48480    2.251  0.0250 *
## Cards        18.21190     4.31865    4.217 3.08e-05 ***
## Age          -0.62406     0.29182   -2.139  0.0331 *
## StudentYes   425.60994    16.50956   25.780  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 98.61 on 393 degrees of freedom
## Multiple R-squared:  0.9547, Adjusted R-squared:  0.954
## F-statistic: 1380 on 6 and 393 DF, p-value: < 2.2e-16
```

```
# Calculate VIF
library(car)
vif_values <- vif(credit_model_reduced)
# If vif() returns a matrix (e.g., for multi-df terms),
```



```

# use the GVIF^(1/(2*Df)) column
if (is.matrix(vif_values)) {
  vif_clean <- data.frame(Variable = rownames(vif_values),
                          VIF = vif_values[, "GVIF^(1/(2*Df))"])
} else {
  # If vif() returns a named numeric vector, use it directly
  vif_clean <- data.frame(Variable = names(vif_values),
                          VIF = as.numeric(vif_values))
}
# Print clean VIF table
print(vif_clean, row.names = FALSE)

```

```

## Variable      VIF
## Income    2.776906
## Limit 229.238479
## Rating 230.869514
## Cards    1.439007
## Age      1.039696
## Student   1.009064

```

7. (10 marks) Answer the questions below by referring to Case 7.

The Chief Risk Officer wants to know whether the model is strong enough to deploy for credit-limit decisions. What is the **proportion of variance in Balance explained** by the model? (Round to **2 decimals**.)

The portfolio team wants a conservative “typical error band” around predicted balances for monitoring drift. What is the model’s **typical prediction error scale**? (Round to **2 decimals**.)

To make the model more stable, the analytics team must identify the variable that is most likely to cause **unstable coefficient estimates** due to multicollinearity. Which predictor(s) should be investigated?

The bank is considering a targeted student credit product, but wants to quantify the incremental balance effect attributable to student status *after controlling for*

income, limit, rating, cards, and age. What is the **estimated incremental effect** of being a student?

Senior management asks: "Which test statistic in the regression output provides the evidence that at least one predictor improves the model beyond an intercept-only baseline?" Which statistic would you report?

Case 8: OmniRetail 360 is a fast-growing omnichannel company (app + stores) selling groceries, personal care, and home essentials. Over the last 6 months, the CEO has pushed three business goals:

1. **Reduce churn** in the loyalty program (customers who stop purchasing for 60+ days).
2. **Improve targeted promotions** without damaging margins (avoid offering discounts to customers who would buy anyway).
3. **Increase operational efficiency** by using fewer features/models while preserving decision quality.

The analytics team has built a pipeline:

- A **churn prediction model** (supervised classification) trained on customer history.
- A **segmentation model** to group customers by buying patterns (unsupervised learning).
- A **campaign-response model** to decide whom to target (supervised learning under imbalance).

However, the board has raised governance questions:

- "Are we accidentally overfitting and inflating performance?"
- "Are we choosing the right metric for the business cost of errors?"
- "If some variables are highly correlated (e.g., `CreditLimit` and `SpendingScore`), are our insights reliable?"

- "Can we simplify the model (e.g., PCA / feature reduction) without losing much business signal?"

8. (10 marks) Answer the questions below by referring to Case 8.

The marketing head wants customer groups discovered **without any labels** to design different campaign themes. Which ML task should be used?

The CFO says: "False positives are expensive because they trigger discounts; false negatives are tolerable." Which evaluation metric should be prioritized for the campaign-response model?

During stress testing, the team notices performance drops sharply on new regions: the model fits training well but does not generalize. Which problem is most likely occurring?

The CTO wants to compress 20 behavioral variables into 2 or 3 dimensions for visualization while keeping most information. Which task should be used?

The analytics team must estimate performance reliably on limited data while using almost every observation for training. Which validation method should be used?

Space for Rough Work

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